Activity 2

Brightlearn
SQL Fundamentals
Aggregate Functions & Grouping

1 Count () Function
Write a SQL query to find the total number of employees in the company

```
Select count (id) AS number of employees
From employees_db.employees;
```

number of employees
10

2 SUM () Function
Write a SQL query to find the total salary paid to all employees in the IT department

```
Select SUM(Salary),
       department
FROM employees_db.employees
WHERE department = IT;
```

Salary	Department
280 000	IT

3 AVG() Function

Write a SQL query to calculate the average salary of employees in the HR department

```
Select AVG(salary) AS Average salary
FROM employees_db.employee
WHERE department = HR
```

department	Average Salary
HR	49 500

4 MIN() and MAX() function

Write a SQL query to find the highest and lowest salary in the company

```
Select MIN(salary) AS lowest salary
       MAX(salary) AS highest salary
FROM employees_db.employee;
```

lowest Salary	highest salary
48 000	62 000

5 Group By Statement

Write a SQL query to ~~p~~ group employees by department and display the total salary paid in each department

```
Select  department,
        SUM (salary) AS total salary
FROM    employee_db . employee
GROUP BY department ;
```

Employee

department	total salary
IT	220 000
HR	99 000
Finance	119 000
Marketing	105 000

6 GROUP BY and COUNT( )

write a SQL query to count how employees work in each city

```
Select COUNT ( id ) AS Total employees in city,
       city
FROM employee_db . employee
GROUP BY city ;
```

Employee

Total employee in City	City
2	New York
3	Chicago
2	Los Angeles
2	San Francisco
1	Houston

7 GROUP BY and ORDER BY

Write a SQL query to group employees by department, calculate the average salary in each department and order the results in descending order of average salary

```
Select AVG (salary) AS average salary
       department,
       COUNT(id) AS total Employees
FROM employee_db . employee,
GROUP BY department
ORDER BY DSC average salary
```

IT 55 000

HR 49 500

Finan 59 500

mark 59 500

Employees

Average Salary	Department	total employees
59 500	Finance	2
55 000	IT	4
52 500	Marketing	2
49 500	HR	2

8 Having Clause
Write SQL query to find departments where the total salary paid exceeds 100,000. (USE GROUP BY and HAVING)

```
Select SUM(salary), AS total salary
       department
FROM employee_db.employee
GROUP BY department
HAVING total salary > 100 000;
```

Employee

Total Salary	Department
220 000	IT
119 000	Finance
105	Marketing

9 Combining Group By, HAVING and ORDER BY
Write a SQL query to list cities where more than one employee works, ordered by the number of employees in descending order

```
Select COUNT(id) As total employees
       City
FROM employee_db.employee
WHERE total employee > 2
ORDER BY DSC total employees
```

Employee

Total Employee	Cities
3	Chicago
2	New York
2	Los Angeles
2	San Francisco
1	Houston

10 Combining Aggregate Functions
Write a SQL query to find the department with the highest average salary

```
Select AVG (salary) AS Average Salary,
       department
FROM   employee_db .employee
GROUP BY department,
ORDER BY DSC avarage salary
Limit 1;
```

Average Salary	Department
59500	Finance